

POINT MASS Mechanical entity with negligible dimensions w.r.t. the environment where it moves -

A point mass (or an object) in general is isolated if there is no interaction with other entity

FORCE Mechanical entity responsible for the interaction among bodies -

Forces are vectors and are absolute (i.e. they are the same for all the observers) (*)

A point mass is isolated if it is not subject to forces

I Newton's Law An isolated point mass has constant velocity w.r.t. any inertial ref. frame

II Newton's Law Given an inertial ref. frame $\langle O \rangle$

$$m \underline{a} = \underline{F}$$

$\uparrow$ acc. of the point $\uparrow$ resultant of the forces

$\underline{a} \equiv \underline{a}_P$

III Newton's Law Given two points P and Q the force of P on Q $\underline{F}_{P \rightarrow Q} = - \underline{F}_{Q \rightarrow P}$ and $\underline{F}_{P \rightarrow Q} \times (\underline{P} - \underline{Q}) = 0$

NOTE Newton's $\underline{n}$ Law can be reformulated as a ^{13/} set of $\underline{n}$ order diff. equations:

$$m \underline{\ddot{x}}(t) = \underline{F}(\underline{x}, \underline{\dot{x}}, t)$$

Fundamental problems

- Direct dynamic problem "compute" the solution $\underline{x} = \underline{x}(t)$ given $\underline{F}$ and $\underline{x}(0), \underline{\dot{x}}(0)$
- Inverse dynamic problem "compute" the force $\underline{F}$ required to obtain a "desired" motion $\underline{x}^* = \underline{x}^*(t)$ -

NOTE The control problem is a "special" case of the inverse dynamic problem.

NOTE The direct dynamic problem in computational terms corresponds to the simulation -

NOTE $\underline{n}$ IL equations are non-linear (most of the time) 2nd order differential eqs. -

Usually (i.e. except for special/pathological cases) they have a unique solution given initial conditions $\underline{x}(0), \underline{\dot{x}}(0)$

Newton's II Law IN ACCELERATED REF FRAMES (*) 14/

Let $\langle a \rangle$ an accelerated ref frame w.r.t. $\langle 0 \rangle$
Then by "NII L"

$$m \underset{\substack{\uparrow \\ \underline{a}_{P/0}}}{\underline{a}} = \underline{F} \Leftrightarrow m \left[\underline{a}_{P/A} + 2 \underline{\omega}_{A/0} \times \underline{v}_{P/A} + \underline{\dot{\omega}}_{A/0} \times \underline{r}_{P/A} + \underline{\omega}_{A/0} \times (\underline{\omega}_{A/0} \times \underline{r}_{P/A}) \right] = \underline{F}$$

Then from the standpoint of $\langle a \rangle$ who "sees" a (relative) acceleration of P expressed by $\underline{a}_{P/A}$, there are additional terms which can be interpreted by "apparent forces" (**)

$$\underline{F}_c = -m \times 2 \underline{\omega}_{A/0} \times \underline{v}_{P/A} \quad \text{Coriolis force}$$

$$\underline{F}_s = -m \underline{\dot{\omega}}_{A/0} \times \underline{r}_{P/A} - m \underline{\omega}_{A/0} \times (\underline{\omega}_{A/0} \times \underline{r}_{P/A})$$

$\uparrow$
centrifugal force

(**) Apparent since they are not seen by other observers.

(*) A.K.A relative dynamics

EQUILIBRIUM A point mass is in equilibrium w.r.t. ^{generic} observer $\langle a \rangle$ if $\underline{v}_{p/a} \equiv 0 \quad \forall t$ 15/

Equilibrium conditions in inertial frames (STATICS)

$$\underline{v}_{p/a} \equiv 0 \Rightarrow \underline{a}_{p/a} \equiv 0 \Leftrightarrow \underline{F} \equiv 0 \quad \forall t$$

Equilibrium conditions in accelerated frames (RELATIVE STATICS)

$$\underline{v}_{p/a} \equiv 0 \Rightarrow \underline{a}_{p/a} \equiv 0 \Leftrightarrow \underline{F} + \underline{F}_S = 0 \quad (*)$$

ENERGY

A force $\underline{F}$ is conservative if there exists a scalar function $V(\underline{x})$ s.t.

$$\underline{F} = - \nabla V$$

$\uparrow$
 $\frac{\partial}{\partial \underline{x}} V(\underline{x})$

$V(\underline{x})$ potential energy (is a function of the position)

$$T = \frac{1}{2} m \underline{\dot{x}} \cdot \underline{\dot{x}}$$

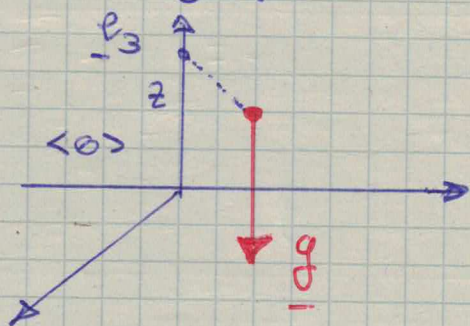
Kinetic energy

(*) Coriolis force must be zero at relative eq. since $\underline{v}_{p/a} \equiv 0$

Examples of conservative forces

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Gravity (near earth)



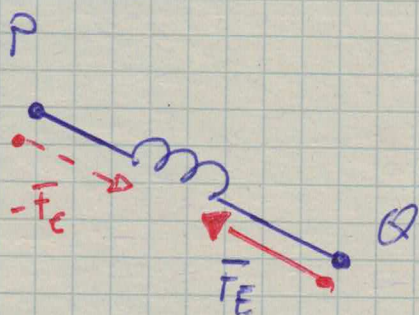
$$\vec{F}_g = -g \underline{e}_3$$

$$9.81 \text{ m/sec}^2$$

$$V_g = mg (\underline{x} \cdot \underline{e}_3) \underline{e}_3$$

$$\underline{z}$$

Elastic force (between two points)



$$\vec{F}_{-E} = -K_E (\underline{Q} - \underline{P}) = -K_E \frac{\underline{z}}{|\underline{z}|}$$

$$V_E = \frac{1}{2} K_E (\underline{z} \cdot \underline{z})$$

In general all constant forces and central forces (e.g. gravity) are conservative

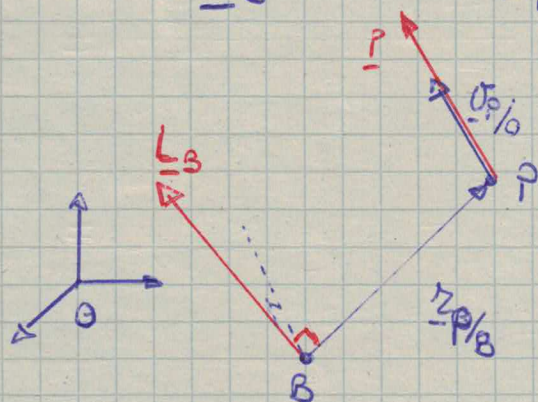
Linear Momentum

$$\underline{p} = m \underline{v}_p$$

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Angular momentum (about a point)

$$\underline{L}_B = m \underline{r}_{p/B} \times \underline{v}_{p/o}$$



Momentum of a force (about a point)

$$\underline{m} = \underline{r}_{p/B} \times \underline{F}$$

